

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12414348
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Ikura; Yoshihiro

Insulated gate bipolar transistor

Abstract

An insulated gate bipolar transistor includes: a gate electrode embedded in a gate trench through a gate insulating film, the gate trench penetrating an emitter region and a base region; and a dummy electrode embedded in a dummy trench through a dummy insulating film, the dummy trench penetrating the emitter region and the base region and being disposed on each side of the gate trench and laterally spaced from each side of the gate trench so as to laterally face the gate trench through the base region, wherein the dummy electrode includes a bottom dummy conductive member disposed at a bottom of the dummy trench such that an upper surface of the bottom dummy conductive member is located lower than a lower surface of the base region, the bottom dummy conductive member being configured to be electrically connected to a gate potential.

Inventors:	Ikura; Yoshihiro (Nagano, JP)
Applicant:	Fuji Electric Co., Ltd. (Kanagawa, JP)
Family ID:	85386704
Assignee:	FUJI ELECTRIC CO., LTD. (Kanagawa, JP)
Appl. No.:	17/878812
Filed:	August 01, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230072478 A1	Mar. 09, 2023

Foreign Application Priority Data

JP	2021-146153	Sep. 08, 2021
----	-------------	---------------

Publication Classification

Int. Cl.: H10D64/00 (20250101); H10D12/00 (20250101); H10D62/17 (20250101); H10D84/60 (20250101)

U.S. Cl.:

CPC H10D64/117 (20250101); H10D12/481 (20250101); H10D62/393 (20250101); H10D84/617 (20250101);

Field of Classification Search

CPC: H01L (29/407); H01L (27/0664); H01L (29/1095); H01L (29/7397); H01L (29/0696); H01L (29/42376); H01L (29/4238); H01L (27/0629); H10D (64/117); H10D (12/481); H10D (62/393); H10D (84/617); H10D (62/127); H10D (64/518); H10D (64/519); H10D (84/811)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2018/0323294	12/2017	Okuda et al.	N/A	N/A
2019/0027472	12/2018	Naito	N/A	H01L 29/42376
2019/0088769	12/2018	Matsushita	N/A	N/A
2020/0144403	12/2019	Nishi	N/A	H01L 29/407
2022/0157976	12/2021	Konishi et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2017-147431	12/2016	JP	N/A
2019-057702	12/2018	JP	N/A
2020-77727	12/2019	JP	N/A
2022-078755	12/2021	JP	N/A
2017/126167	12/2016	WO	N/A
2018/074427	12/2017	WO	N/A

OTHER PUBLICATIONS

K. Nishi et al., “CSTBT(TM) Based Split-Gate RC-IGBT with Low Loss and EMI Noise”, Proceedings of the 32nd International Symposium on Power Semiconductor Devices & ICs (ISPSD), Sep. 2020, pp. 138-141 (Mentioned on paragraph Nos. 3-4 and 6 of the application.). cited by applicant

Japanese Office Action dated Apr. 30, 2025 in a counterpart Japanese patent application No. 2021-146153. (A machine translation (not reviewed for accuracy) attached.). cited by applicant

Primary Examiner: Valenzuela; Patricia D

Attorney, Agent or Firm: CHEN YOSHIMURA LLP

Background/Summary

BACKGROUND OF THE INVENTION

Technical Field

(1) The present invention relates to a trench gate type insulated gate bipolar transistor (IGBT).

Background Art

(2) Conventionally, in a trench gate type IGBT, a high withstand voltage can be maintained even if a high-concentration carrier accumulation layer is provided by arranging the gate trench spacing narrowly. In that case, the density of the gate trenches becomes high, and the total gate charge Q_g increases. Therefore, the increase in the total gate charge Q_g is eliminated by forming dummy trenches where some of the gate trenches are connected to the emitter potential. However, by providing the dummy trenches, the potential around the gate trenches is lowered, and the gate-collector capacitance $C_{sub.GC}$ increases during the turn-on of the IGBT, which causes the generation of the collector voltage tail. Therefore, the turn-on loss increases.

(3) Non-Patent Document 1 proposes a split gate structure in which an electrode in a gate trench is divided into an upper part and a lower part, and an upper electrode is connected to a gate potential and a lower electrode is connected to an emitter potential. The split gate structure reduces the gate-collector capacitance $C_{sub.GC}$ and realizes a low-loss IGBT. Further, in Patent Document 1, the electrode of the dummy trench is divided into two, and the upper conductive member is connected to the emitter potential and the lower conductive member is connected to the gate potential to adjust the gate-collector capacitance $C_{sub.GC}$ as well as the collector-emitter capacitance $C_{sub.CE}$.

(4) Since the lower electrode of the gate trench of Non-Patent Document 1 and the upper conductive portion of Patent Document 1 are connected to the emitter potential, it is difficult to perform a screening using the withstand voltage test of the insulating film of the gate trench or dummy trench by applying a voltage. Although it is possible to perform such a screening in the middle of the manufacturing process, that would cause an increase in manufacturing cost. In Patent Document 1, since the gate electrodes are adjacent to each other, the holes accumulated on the upper surface side at the time of turn-on push the potential upwards, and the displacement current flows through the gate electrode. Therefore, the rate of change of the transient current di/dt increases and noise increases.

RELATED ART DOCUMENT

Patent Document

(5) Patent Document 1: International publication No. 2018/074427

Non-Patent Document

(6) Non-Patent Document 1: K. Nishi et al., "CSTB™ Based Split-Gate RC-IGBT with Low Loss and ENE Noise", Proceedings of the 32nd International Symposium on Power Semiconductor Devices & ICs (ISPSD), September 2020, pp. 138-141.

SUMMARY OF THE INVENTION

(7) In view of the above problems, it is an object of the present invention to provide an insulated gate bipolar transistor capable of reducing turn-on loss and for which a screening for a gate insulating film can be performed with relative ease.

(8) Additional or separate features and advantages of the invention will be set forth in the descriptions that follow and in part will be apparent from the description, or may be learned by practice of the invention. The objectives and other advantages of the invention will be realized and attained by the structure particularly pointed out in the written description and claims thereof as well as the appended drawings.

(9) To achieve these and other advantages and in accordance with the purpose of the present invention, as embodied and broadly described, in one aspect, the present disclosure provides an insulated gate bipolar transistor, comprising: a drift layer made of semiconductor of a first conductivity type; a base region made of semiconductor of a second conductivity type on the drift

layer; an emitter region made of semiconductor of the first conductivity type on the base region, an impurity concentration of the emitter region being higher than that in the drift layer; a gate electrode embedded in a gate trench through a gate insulating film, the gate trench penetrating the emitter region and the base region; and a dummy electrode embedded in a dummy trench through a dummy insulating film, the dummy trench penetrating the emitter region and the base region and being disposed on each side of the gate trench and laterally spaced from each side of the gate trench so as to laterally face the gate trench through the base region, wherein the gate electrode is configured to be electrically connected to a gate potential, and wherein the dummy electrode includes a bottom dummy conductive member disposed at a bottom of the dummy trench such that an upper surface of the bottom dummy conductive member is located lower than a lower surface of the base region, the bottom dummy conductive member being configured to be electrically connected to the gate potential.

(10) According to the present invention, it is possible to provide an insulated gate bipolar transistor capable of reducing turn-on loss and for which a screening for a gate insulating film can be performed with relative ease.

(11) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory, and are intended to provide further explanation of the invention as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic plan view showing an example of an IGBT according to a first embodiment of the present invention.

(2) FIG. 2 is a schematic perspective view including a cross section seen from the direction of the line A-A in FIG. 1.

(3) FIG. 3 is a schematic cross-sectional view seen from the direction of the line B-B in FIG. 1.

(4) FIG. 4 is a schematic cross-sectional view seen from the direction of the line C-C in FIG. 1.

(5) FIG. 5 is a diagram showing an example of simulated turn-on waveforms of the IGBT of the first embodiment.

(6) FIG. 6 is a diagram showing an example of a simulated potential distribution around a gate trench during turn-on of the IGBT of the first embodiment.

(7) FIG. 7 is a schematic cross-sectional view showing an example of an IGBT according to a comparative example.

(8) FIG. 8 is a diagram showing an example of simulated turn-on waveforms of an IGBT of the comparative example.

(9) FIG. 9 is a diagram showing an example of simulated potential distribution around a gate trench during the turn-on of the IGBT of the comparative example.

(10) FIG. 10 is a schematic plan view showing an example of an IGBT according to a first modification of the first embodiment.

(11) FIG. 11 is a schematic cross-sectional view seen from the direction of the line D-D in FIG. 10.

(12) FIG. 12 is a schematic cross-sectional view seen from the direction of the line E-E in FIG. 10.

(13) FIG. 13 is a schematic plan view showing an example of an IGBT according to a second modification of the first embodiment.

(14) FIG. 14 is a schematic cross-sectional view seen from the direction of the line F-F in FIG. 13.

(15) FIG. 15 is a schematic cross-sectional view seen from the direction of the line G-G in FIG. 13.

(16) FIG. 16 is a schematic cross-sectional view showing an example of an IGBT according to a second embodiment.

(17) FIG. 17 is a schematic plan view showing an example of an IGBT according to a third

embodiment.

(18) FIG. 18 is a schematic cross-sectional view seen from the direction of the line H-H in FIG. 17.

(19) FIG. 19 is a schematic cross-sectional view seen from the direction of the line I-I in FIG. 17.

(20) FIG. 20 is a schematic cross-sectional view seen from the direction of the line J-J in FIG. 17.

(21) FIG. 21 is a schematic cross-sectional view showing a connection between the trenches and the diode in the IGBT of the third embodiment.

(22) FIG. 22 is a diagram showing an example of simulated turn-on waveforms of the IGBT of the third embodiment.

(23) FIG. 23 is a diagram showing an example of simulated turn-off waveforms of the IGBT of the third embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

(24) Hereinafter, embodiments of the present invention will be described with reference to the drawings. In the description of the drawings referred to in the following description, the same or similar parts are designated by the same or similar reference numerals. However, it should be noted that the drawings are schematic, and the relationship between the thickness and the plane dimensions, the ratio of the thickness of the respective layers, etc., are different from the actual ones. Therefore, the specific thickness and dimensions should be determined in consideration of the following explanation. In addition, it goes without saying that parts having different dimensional relationships and ratios may be included among the various drawings.

(25) Further, the definition of the direction such as up and down in the following description is used merely for convenience of explanation, and does not limit the technical idea of the present invention. For example, if the object is rotated by 90° and observed, the top and bottom are converted to left and right, and if the object is rotated by 180° and observed, the top and bottom are reversed. Further, in the following description, the case where the first conductivity type is the n type and the second conductivity type opposite thereto is the p type will be exemplified. However, the conductivity type may be selected in the opposite relationship, the first conductivity type may be the p type, and the second conductivity type may be the n type. Further, + and - attached to n and p mean that the impurity concentration is relatively high or low, respectively, as compared with the semiconductor regions to which + or - is not added. However, even if the semiconductor regions have the same n and n, it does not mean that the impurity concentrations of the respective semiconductor regions are exactly the same. The following structures of the embodiments and modifications thereof can be manufactured by conventional manufacturing techniques widely available and known in the field of insulated gate bipolar transistors.

First Embodiment

(26) FIG. 1 is a plan view schematically showing trenches (7a, 7b) arranged in the active portion of an IGBT according to a first embodiment of the present invention. As shown in FIG. 1, the trenches (7a, 7b) are composed of striped gate trenches 7a and striped dummy trenches 7b arranged side by side on both sides of the gate trench 7a in a plan view. In FIG. 1, gate trenches 7a and dummy trenches 7b are alternately arranged side by side, but the present invention is not limited to this. For example, one or more dummy trenches 7b may be arranged side by side between adjacent two gate trenches 7a. The gate trench 7a has an upper gate conductive member 11a connected to a gate surface electrode 16 via a wiring layer 12a and a contact hole 13a. A bottom gate conductive member 9a divided from the upper gate conductive member 11a by a split insulating film 10a is provided at the tip of the gate trench 7a in the depth direction. The dummy trench 7b has an upper dummy conductive member 11b connected to an emitter surface electrode 15 via a wiring layer 12b and a contact hole 13b. A bottom dummy conductive member 9b divided from an upper dummy conductive member 11b by a split insulating film 10b is provided at the tip of the dummy trench 7b in the depth direction. The bottom gate conductive member 9a and the bottom dummy conductive member 9b are connected to the gate surface electrode 16 via a wiring layer 12c and a contact hole 13c. Although not shown, a gate pad electrically connected to an external gate drive circuit or the

like may be arranged at the end of the active portion. Further, an outer peripheral portion having a high voltage withstand structure may be provided around the active portion.

(27) FIG. 2 is a view of a cross section cut in a direction orthogonal to the direction in which the trenches (7a, 7b) extend in parallel, as viewed from an oblique direction. As shown in FIG. 2, a second conductive type (p type) base region 5 is arranged on a first conductive type (n.sup.- type) drift layer 3. An n.sup.+ type emitter region 6 having a higher impurity concentration than the drift layer 3 is provided above the base region 5. An n-type accumulation layer 4 having a higher impurity concentration than the drift layer 3 is provided below the base region 5. Trenches (7a, 7b) penetrating the base region 5 and the accumulation layer 4 from the upper surface of the emitter region 6 are provided. The side surface of each of the trenches (7a, 7b) is in contact with the emitter region 6, the base region 5, and the accumulation layer 4, and is further in contact with a part of the drift layer 3. The trenches (7a, 7b) penetrate the base region 5 and the accumulation layer 4 from the emitter region 6 to reach the drift layer 3. The dummy trench 7b is provided on both sides of the gate trench 7a so as to face the gate trench 7a with the emitter region 6, the base region 5, and the accumulation layer 4 interposed therebetween. A p.sup.+ type contact region 25 having a higher impurity concentration than the base region 5 is also provided above the base region 5. The contact region 25 is provided alternately with the emitter region 6 in the direction in which the trenches (7a, 7b) extend in parallel.

(28) As shown in FIG. 2, a gate insulating film 8a is provided on the bottom surface and the side surface of the gate trench 7a. The split-type gate electrode (9a, 11a) is embedded inside the gate trench 7a via the gate insulating film 8a. The gate electrode (9a, 11a) is composed of the bottom gate conductive member 9a provided at the bottom of the gate trench 7a and the upper gate conductive member 11a provided on the bottom gate conductive member 9a via the split insulating film 10a. Further, a dummy insulating film 8b is provided on the bottom surface and the side surface of the dummy trench 7b. The split-type dummy electrode (9b, 11b) is embedded inside the dummy trench 7b via the dummy insulating film 8b. The dummy electrode (9b, 11b) is composed of the bottom dummy conductive member 9b provided at the bottom of the dummy trench 7b and the upper dummy conductive member 11b provided on the bottom dummy conductive member 9b via the split insulating film 10b. The lower surfaces of the upper gate conductive member 11a and the upper dummy conductive member 11b are located below the level of the lower surface of the base region 5. The upper surfaces of the bottom gate conductive member 9a and the bottom dummy conductive member 9b are located below the level of the upper surface of the drift layer 3.

(29) As the gate insulating film 8a, in addition to a silicon dioxide (SiO.sub.2) film, a single layer made of one of silicon oxynitride (SiON) film, strontium oxide (SrO) film, silicon nitride (Si.sub.3N.sub.4) film, aluminum oxide (Al.sub.2O.sub.3) film, magnesium oxide (MgO) film, yttrium oxide (Y.sub.2O.sub.3) film, hafnium oxide (HfO.sub.2) film, zirconium oxide (ZrO.sub.2) film, tantalum oxide (Ta.sub.2O.sub.5) film, and bismuth oxide (Bi.sub.2O.sub.3) film, or a composite film made of laminating these films may be adopted. As the split insulating films 10a and 10b, a tetraethoxysilane (TEOS) oxide film, a high resistance polysilicon film, or the like can be used. As the material of the gate electrode (9a, 11a) and the dummy electrode (9b, 11b), a polysilicon layer (doped polysilicon layer) in which impurities such as phosphorus (P) and boron (B) are added to a high impurity concentration, for example, can be used.

(30) Interlayer insulating films 14 are arranged on the upper gate conductive member 11a and the upper dummy conductive member 11b, respectively. The emitter surface electrode 15 is provided so as to cover the interlayer insulating film 14. The emitter surface electrode 15 physically contacts the exposed emitter region 6 between the interlayer insulating films 14. As the interlayer insulating film 14, a silicon oxide film to which boron (B) and phosphorus (P) are added (BPSG) is used. The interlayer insulating film 14 may be a silicon oxide film to which phosphorus (P) is added (PSG), a non-doped SiO.sub.2 film called "NSG" that does not contain phosphorus (P) or boron (B), a silicon oxide film to which boron (B) is added (BSG), Si.sub.3N.sub.4 film, or the like instead.

Further, a laminated film of these materials may be used. The emitter surface electrode **15** can be composed of, for example, a nickel silicide (NiSi.sub.x) film, a titanium nitride (TiN) film, a titanium (Ti) film, an aluminum (Al) film, or an aluminum-silicon (Al—Si) film.

(31) An n.sup.+ type field stop layer (FS layer) **2** is arranged on the lower surface of the drift layer **3**, and a p.sup.+ type collector region **1** is arranged on the lower surface of the FS layer **2**. A collector back surface electrode **17** is arranged on the lower surface of the collector region **1**. As the collector back surface electrode **17**, for example, a single-layer film made of gold (Au) or a metal film laminated in the order of Ti, nickel (Ni), and Au can be used.

(32) FIGS. **3** and **4** are diagrams schematically showing cross sections of the gate trench **7a** and the dummy trench **7b** cut in the extending direction, respectively. In FIGS. **3** and **4**, the emitter region **6**, the contact region **25**, the base region **5**, and the accumulation layer **4** do not appear in these cross sections. As shown in FIG. **3**, the upper gate conductive member **11a** and the bottom gate conductive member **9a** are insulated by the split insulating film **10a**. A wiring layer **12a** is provided on the upper surface of the upper gate conductive member **11a**. A wiring layer **12c** is provided on the upper surface of the bottom gate conductive member **9a** exposed at one end of the gate trench **7a** in the stretching direction and on the field insulating film **21** provided on the drift layer **3**. The wiring layers **12a** and **12c** are connected to the gate surface electrode **16**, which is electrically connected to the gate potential, via the contact holes **13a** and **13c**, respectively. Further, as shown in FIG. **4**, the upper dummy conductive member **11b** and the bottom dummy conductive member **9b** are insulated by the split insulating film **10b**. A wiring layer **12b** is provided on the upper surface of the upper dummy conductive member **11b**. A wiring layer **12c** is provided on the upper surface of the bottom dummy conductive member **9b** exposed at one end of the dummy trench **7b** in the stretching direction and on the field insulating film **21** on the drift layer **3**. The wiring layer **12b** is connected to the emitter surface electrode **15**, which is electrically connected to the emitter potential, via the contact hole **13b**. The wiring layer **12c** is connected to the gate surface electrode **16**, which is electrically connected to the gate potential, via the contact hole **13c**.

(33) During the operation of the IGBT of the first embodiment, for example, a positive voltage is applied to the collector back electrode **17** with the emitter front electrode **15** being at the ground potential. When a positive voltage equal to or higher than the threshold value is applied to the gate electrodes (**9a**, **11a**), an inversion layer (channel) is formed on the side surface of the gate trench **7a** in the base region **5**, and the IGBT turns on. The inversion layer is formed on the surface of the base region **5** in contact with the side surface of the gate trench **7a**, which is the interface between the gate insulating film **8a** sandwiched at the position where the base region **5** faces the upper gate conductive member **11a** and the base region **5**. In the on state, a current flows from the collector back surface electrode **17** to the emitter surface electrode **15** via the collector region **1**, the FS layer **2**, the drift layer **3**, the accumulation layer **4**, the inversion layer in the base region **5**, and the emitter region **6**. When the voltage applied to the gate electrodes (**9a**, **11a**) is less than the threshold value, no current flows from the collector back surface electrode **17** to the emitter front surface electrode **15** because the inversion layer is not formed in the base region **5**.

(34) In the IGBT according to the first embodiment, since the upper dummy conductive member **11b** of the dummy trench **7b** is electrically connected to the emitter potential, an inversion layer is not formed on the surface of the base region **5** in contact with the side surface of the dummy trench **7b**. Therefore, it is possible to reduce the density of the gate trenches **7a** in which the channel of the IGBT is formed, and it is possible to suppress an increase in the total gate charge Qg. Further, a split-type electrode structure is embedded in each of the trenches (**7a**, **7b**). Both the bottom gate conductive member **9a** of the gate trench **7a** and the bottom dummy conductive member **9b** of the dummy trench **7b** are electrically connected to the gate potential. Therefore, it is possible to easily perform an insulation withstand voltage inspection screening on the gate insulating film **8a** and the dummy insulating film **8b** provided on the bottom surface of the trench (**7a**, **7b**) where electric field concentration is likely to occur.

(35) The turn-on characteristics of the IGBT sample of the first embodiment were evaluated. FIG. 5 show simulated turn-on waveforms of the sample. As shown in FIG. 5, when the gate voltage $V_{sub.GE}$ between the gate and the emitter is applied to activate the IGBT sample, the gate current $I_{sub.G}$ first flows, the capacitance between the gate and the emitter is charged, and the $V_{sub.GE}$ rises. When the $V_{sub.GE}$ becomes equal to or higher than the threshold voltage, the collector voltage $V_{sub.CE}$ between the collector and the emitter drops, and the collector current $I_{sub.C}$ starts to flow. When the collector current $I_{sub.C}$ has a low current of 15 A, the time change rate dv/dt of the collector voltage $V_{sub.CE}$ is almost constant and smoothly decreases. Even when the collector current $I_{sub.C}$ is 150 A, the time change rate dv/dt of the collector voltage $V_{sub.CE}$ is almost constant. FIG. 6 shows simulated potential distributions with respect to the distance in the depth direction from the upper surface of the emitter region 6 to the bottom of the gate trench 7a around the gate trench 7a when the collector voltage $V_{sub.CE}$ decreases at intervals of 50 V. As shown in FIG. 6, the potential rises from the junction region between the emitter region 6 and the base region 5 and becomes substantially flat in the accumulation layer 4, and increases near the bottom of the gate trench 7a in the drift layer 3. When the collector voltage $V_{sub.CE}$ decreases, the potential near the bottom of the gate trench 7a decreases, but it does not become lower than the gate voltage $V_{sub.GE}$. Further, no significant change was observed in the potentials around the emitter region 6, the base region 5, and the accumulation layer 4, and the potentials there were almost fixed.

(36) FIG. 7 shows, as a comparative example, a conventional IGBT having a trench (7a, 7b) in which a unitary electrode structure is embedded. As shown in FIG. 7, a unitary gate electrode 11c is provided inside the gate trench 7a via a gate insulating film 8a, and a unitary dummy electrode 11d is provided inside the dummy trench 7b via a dummy insulating film 8b. FIG. 8 shows simulated turn-on waveforms of the comparative example. As shown in FIG. 8, when the IGBT of the comparative example is activated and the gate voltage $V_{sub.GE}$ becomes equal to or higher than the threshold voltage, the collector voltage $V_{sub.CE}$ drops and the collector current $I_{sub.C}$ starts to flow. When the $I_{sub.C}$ has a low current of 15 A, the time change rate dv/dt of the collector voltage $V_{sub.CE}$ is almost constant and smoothly decreases even in the comparative example. On the other hand, when the $I_{sub.C}$ is 150 A, the time change rate dv/dt of the collector voltage $V_{sub.CE}$ is not constant in the comparative example, and a voltage tail T is generated. The cause of the voltage tail T is that the gate-collector capacitance $C_{sub.GC}$ increases transiently. As shown in FIG. 8, since the collector current $I_{sub.C}$ having a high collector current flows during the period of the voltage tail T, the turn-on loss increases.

(37) FIG. 9 shows simulated potential distributions around the gate trench 7a when the collector voltage $V_{sub.CE}$ decreases at intervals of 50 V. As shown in FIG. 9, the potential distribution of the comparative example also increases near the bottom of the gate trench 7a in the same manner as the potential distribution of the example shown in FIG. 6, but the potential value is lower than that of the embodiment example. In the comparative example, the dummy electrode 11d of the dummy trench 7b arranged next to the gate trench 7a is connected to the emitter potential, and this pushes down the potential near the bottom of the gate trench 7a. Further, as shown in FIG. 9, when the collector voltage $V_{sub.CE}$ drops from 550V to 300V, mainly the potential near the bottom of the gate trench 7a drops, but it does not become lower than the gate voltage $V_{sub.GE}$. In FIG. 9, when the $V_{sub.CE}$ is in the range P of 250 V to 50 V, as the $V_{sub.CE}$ descends, the potential near the bottom of the gate trench 7a becomes lower than that of the $V_{sub.GE}$, and further, the potential drop also occurs in the n-type semiconductor region from the drift layer 3 to the accumulation layer 4. Such a large decrease in potential in the n-type semiconductor region corresponds to an increase in the gate-collector capacitance $C_{sub.GC}$. That is, as shown in FIG. 8, since the gate-collector capacitance $C_{sub.GC}$ is charged with a substantially constant gate current $I_{sub.G}$, the time required for charging increases and the voltage tail T is generated.

(38) As described above, in the first embodiment, the trenches (7a, 7b) are each embedded with a

split-type electrode structure. Both the upper gate conductive member **11a** and the bottom gate conductive member **9a** of the gate trench **7a** are electrically connected to the gate potential. The upper dummy conductive member **11b** of the dummy trench **7b** is electrically connected to the emitter potential, while the bottom dummy conductive member **9b** is electrically connected to the gate potential. Therefore, as the collector voltage $V_{sub,CE}$ decreases, the potential at the bottom of the gate trench **7a** also decreases, but the potential near the bottom of the gate trench **7a** is raised higher than the gate voltage $V_{sub,GE}$. As a result, it is possible to suppress a decrease in potential in the n-type semiconductor region from the drift layer **3** to the accumulation layer **4**, and it is possible to prevent the generation of the voltage tail T in the collector voltage $V_{sub,CE}$. Thus, in the IGBT according to the first embodiment, the turn-on loss can be reduced and a screening on the gate insulating film can be performed with ease.

(39) (First Modification)

(40) As shown in FIG. **10**, the IGBT according to a first modification of the first embodiment of the present invention includes striped trenches (**7a**, **7b**) arranged side by side in a plan view. The gate trench **7a** has a unitary gate electrode **11c** connected to the gate surface electrode **16** via a wiring layer **12c** and a contact hole **13c**. The dummy trench **7b** has the upper dummy conductive member **11b** and the bottom dummy conductive member **9b** divided by the split insulating film **10b**. The upper dummy conductive member **11b** is connected to the emitter surface electrode **15** via the wiring layer **12b** and the contact hole **13b**. The bottom dummy conductive member **9b** is connected to the gate surface electrode **16** via the wiring layer **12c** and the contact hole **13c**, as in the first embodiment. As shown in FIG. **11**, the trench (**7a**, **7b**) penetrates the base region **5** and the accumulation layer **4** from the emitter region **6** and reaches the drift layer **3**. The gate electrode **11c** is embedded inside the gate trench **7a** via a gate insulating film **8a**. The split-type dummy electrodes (**9b**, **11b**) are embedded inside the dummy trench **7b** via the dummy insulating film **8b**. The interlayer insulating films **14** are arranged on the gate electrode **11c** and the upper dummy conductive member **11b**, respectively. The first modification is different from the first embodiment in that the unitary gate electrode **11c** is embedded in the gate trench **7a**. Since the other configurations of the IGBT of the first modification are the same as those of the first embodiment, duplicated description will be omitted.

(41) As shown in FIG. **12**, at one end of the gate trench **7a** in the extending direction, the wiring layer **12c** is provided in physical contact with the gate electrode **11c** and the upper surface of the field insulating film **21**. The gate electrode **11c** is connected to the gate surface electrode **16** electrically connected to the gate potential via the wiring layer **12c** and the contact hole **13c**. The gate electrode **11c** embedded in the gate trench **7a** is a unitary type, which simplifies the manufacturing process of the gate electrode structure. As a result, potential manufacturing damages to the gate insulating film **8a** can be reduced.

(42) Further, as shown in FIG. **4**, the bottom dummy conductive member **9b** embedded in the inner bottom of the dummy trench **7b** is connected the gate surface electrode **16**, which is electrically connected to the gate potential, via the wiring layer **12c** and the contact hole **13c**. Therefore, it is possible to easily perform an insulation withstand voltage inspection screening on the gate insulating film **8a** and the dummy insulating film **8b** provided on the bottom surface of the trench (**7a**, **7b**) where electric field concentration is likely to occur.

(43) Further, in the IGBT according to the first modification, the upper dummy conductive member **11b** of the dummy trench **7b** is electrically connected to the emitter potential as in the IGBT of the first embodiment shown in FIG. **4**. Therefore, the inversion layer is not formed on the surface of the base region **5** in contact with the side surface of the dummy trench **7b**. Therefore, the density of the gate trenches **7a** on which the IGBT channel is formed is reduced, and it is possible to suppress an increase in the total gate charge Q_g .

(44) Further, although the upper dummy conductive member **11b** of the dummy trench **7b** is electrically connected to the emitter potential, the bottom dummy conductive member **9b** is

electrically connected to the gate potential. Therefore, even if the collector voltage $V_{sub.CE}$ decreases, the potential near the bottom of the gate trench **7a** is raised higher than the gate voltage $V_{sub.GE}$. As a result, it is possible to suppress a decrease in potential in the n-type semiconductor region from the drift layer **3** to the accumulation layer **4**, and it is possible to prevent the generation of a voltage tail in the collector voltage $V_{sub.CE}$. Thus, the turn-on loss can be reduced also in the IGBT of the first modification.

(45) (Second Modification)

(46) As shown in FIG. **13**, the IGBT according to a second modification of the first embodiment of the present invention includes striped trenches (**7a**, **7b**) arranged side by side in a plan view. The gate electrode **11c** of the gate trench **7a** is connected to the gate surface electrode **16** via the wiring layer **12c** and the contact hole **13c**. The dummy trench **7b** has an upper embedded insulating film **10c** and a bottom dummy conductive member **9b**. The bottom dummy conductive member **9b** is connected to the gate surface electrode **16** via the wiring layer **12c** and the contact hole **13c**. As shown in FIG. **14**, the trench (**7a**, **7b**) penetrates the base region **5** and the accumulation layer **4** from the emitter region **6** and reaches the drift layer **3**. A unitary gate electrode **11c** is embedded inside the gate trench **7a** via a gate insulating film **8a**. Inside the dummy trench **7b**, the upper embedded insulating film **10c** is embedded in the upper part and the bottom dummy conductive member **9b** is embedded under the upper embedded insulating film **10c** via the dummy insulating film **8b**. The lower surface of the upper embedded insulating film **10c** is located below the level of the lower surface of the base region **5**. As the upper embedded insulating film **10c**, a TEOS oxide film, a high resistance polysilicon film, or the like can be used. The interlayer insulating films **14** are arranged on the gate electrode **11c** and the upper embedded insulating film **10c**, respectively. The second modification is different from the first modification in that the upper embedded insulating film **10c** and the bottom dummy conductive member **9b** are embedded in the dummy trench **7b**. Since the other configurations of the IGBT of the second modification are the same as those of the first modification, duplicate description will be omitted.

(47) As shown in FIG. **15**, at one end of the dummy trench **7b** in the extending direction, the wiring layer **12c** is provided in physical contact with the bottom dummy conductive member **9b** and the upper surface of the field insulating film **21**. The bottom dummy conductive member **9b** is connected to the gate surface electrode **16**, which is electrically connected to the gate potential, via the wiring layer **12c** and the contact hole **13c**. The gate trench **7a** is the same as that of the first modification shown in FIG. **12**, and the embedded unitary gate electrode **11c** is electrically connected to the gate potential. Therefore, the manufacturing process of the gate electrode structure is simplified, and potential manufacturing damage to the gate insulating film **8a** can be reduced. Further, it becomes possible to easily perform an insulation withstand voltage inspection screening on the gate insulating film **8a** and the dummy insulating film **8b** provided on the bottom surface of the trench (**7a**, **7b**) where electric field concentration is likely to occur.

(48) Further, in the IGBT according to the second modification, the upper embedded insulating film **10c** of the dummy trench **7b** has a floating potential. Therefore, the inversion layer is not formed on the surface of the base region **5** in contact with the side surface of the dummy trench **7b**. Therefore, the density of the gate trenches **7a** on which the IGBT channel is formed is reduced, and it is possible to suppress an increase in the total gate charge Q_g . Further, the bottom dummy conductive member **9b** is electrically connected to the gate potential. Therefore, even if the collector voltage $V_{sub.CE}$ decreases, the potential near the bottom of the gate trench **7a** is raised higher than the gate voltage $V_{sub.GE}$. As a result, it is possible to suppress a decrease in potential in the n-type semiconductor region from the drift layer **3** to the accumulation layer **4**, and it is possible to prevent the generation of a voltage tail in the collector voltage $V_{sub.CE}$. Thus, the turn-on loss can be reduced also in the IGBT of the second modification.

Second Embodiment

(49) As shown in FIG. **16**, the IGBT according to a second embodiment of the present invention

includes trenches (7a, 7b) arranged side by side, similarly to the first modification of the first embodiment. The gate electrode 11c is embedded inside the gate trench 7a via the gate insulating film 8a, and the split-type dummy electrodes (9b, 11b) are embedded inside the dummy trench 7b via the dummy insulating film 8b. In the second embodiment, as shown in FIG. 16, a p-type trench bottom floating layer 20 is provided above the n-type drift layer 3. The trenches (7a, 7b) penetrate the base region 5 and the accumulation layer 4 from the emitter region 6 and reach the trench bottom floating layer 20. The upper surface of the trench bottom floating layer 20 is located above the level of the lower surface of the upper dummy conductive member 11b of the dummy trench 7b by a distance S. That is, the trench bottom floating layer 20 and the upper dummy conductive member 11b overlap each other by the distance S. Here, instead of the unitary gate electrode 11c, the split-type gate electrodes (9a, 11a) shown in FIGS. 2 and 3 may be embedded in the gate trench 7a. The second embodiment is different from the first modification of the first embodiment in that the trench bottom floating layer 20 is provided on the upper part of the drift layer 3. Since the other configurations of the IGBT of the second embodiment are the same as those of the first modification of the first embodiment, duplicate description will be omitted.

(50) Without the p-type trench bottom floating layer, as shown in FIG. 5, the time change rate dv/dt of the collector voltage $V_{sub,CE}$ at turn-on becomes slower as the collector current $I_{sub,C}$ becomes larger. In order to suppress the generation of noise, it is necessary to suppress the time change rate dv/dt . If the time change rate dv/dt when the collector current $I_{sub,C}$ is small is regulated to a specified value, the time change rate dv/dt is suppressed further when the collector current $I_{sub,C}$ is large, and the turn-on loss increases. In the second embodiment, the p-type trench bottom floating layer 20 that overlaps the upper dummy conductive member 11b of the dummy trench 7b by the distance S is provided. At the time of turn-on, holes are accumulated around the upper dummy conductive member 11b electrically connected to the emitter potential, and the trench bottom floating layer 20 is electrically connected to the emitter potential via the accumulated holes. As a result, the gate-collector capacitance $C_{sub,GC}$ becomes constant, and the dependence of the time change rate dv/dt on the collector current $I_{sub,C}$ is suppressed.

(51) As described above, according to the IGBT of the second embodiment, it is possible to reduce the collector current $I_{sub,C}$ dependence of the time change rate dv/dt of the collector voltage $V_{sub,CE}$ at the time of turn-on, and it is possible to further suppress an increase in turn-on loss. The other effects of the IGBT of the second embodiment are the same as those of the IGBT of the first modification of the first embodiment. In the second embodiment, as described above, the unitary gate electrode 11c is used as the electrode structure embedded in the gate trench 7a, but the present invention is not limited to this. For example, as the electrode structure to be embedded in the gate trench 7a, the split-type gate electrodes (9a, 11a) shown in FIG. 2 may be used.

Third Embodiment

(52) In the IGBT according to a third embodiment of the present invention, as shown in FIG. 17, striped trenches (7a, 7b) are arranged side by side alternately in a plan view similarly to the second modification of the first embodiment. A dummy surface electrode 18 is arranged between the emitter surface electrode 15 and the gate surface electrode 16a, and a gate electrode pad 19 is arranged apart from the gate surface electrode 16a. The gate electrode 11c of the gate trench 7a is connected to the gate surface electrode 16a via the wiring layer 12d and the contact hole 13d. The bottom dummy conductive member 9b of the dummy trench 7b is connected to the dummy surface electrode 18 via the wiring layer 12e and the contact hole 13e. As shown in FIG. 18, an unitary gate electrode 11c is embedded inside the gate trench 7a via the gate insulating film 8a. Inside the dummy trench 7b, the upper embedded insulating film 10c is embedded in the upper part and the bottom dummy conductive member 9b is embedded under the upper embedded insulating film 10c via the dummy insulating film 8b. Further, as shown in FIG. 17, the IGBT according to the third embodiment includes a diode 22 and a resistance element 24. The diode 22 is arranged between the dummy surface electrode 18 and the gate surface electrode 16a for each of the dummy trenches 7b.

The resistance element **24** is arranged between the gate surface electrode **16a** and the gate electrode pad **19**. An anode region **22a** of the diode **22** is electrically connected to the gate surface electrode **16a** via a contact hole **23a**. A cathode region **22b** of the diode **22** is electrically connected to the dummy surface electrode **18** via a contact hole **23b**. One end of the resistance element **24** is electrically connected to the gate surface electrode **16a** via a contact hole **23c**, and the other end is electrically connected to the gate electrode pad **19** via a contact hole **23d**. Since the other configurations of the IGBT according to the third embodiment are the same as those of the second modification of the first embodiment, duplicate description will be omitted.

(53) As shown in FIG. **19**, at one end of the gate trench **7a** in the extending direction, the wiring layer **12d** is provided in physical contact with the gate electrode **11c** and the upper surface of the field insulating film **21**. An interlayer insulating film **14** is composed of a first interlayer insulating film **14a** that covers the wiring layer **12d** and a second interlayer insulating film **14b** provided on the first interlayer insulating film **14a**. The wiring layer **12d** is electrically connected to the gate surface electrode **16a** via a contact hole **13d** penetrating the first and second interlayer insulating films **14a** and **14b**. As shown in FIG. **20**, at one end of the dummy trench **7b** in the extending direction, the wiring layer **12e** is provided in physical contact with the bottom dummy conductive member **9b** and the upper surface of the field insulating film **21**. The bottom dummy conductive member **9b** is electrically connected to the dummy surface electrode **18** via a contact hole **13e** penetrating the first and second interlayer insulating films **14a** and **14b**.

(54) As shown in FIG. **20**, the diode **22** is provided between the first and second interlayer insulating films **14a** and **14b**. The anode region **22a** of the diode **22** is electrically connected to the gate surface electrode **16a** via the contact hole **23a** provided in the second interlayer insulating film **14b**. The cathode region **22b** of the diode **22** is electrically connected to the dummy surface electrode **18** via the contact hole **23b** provided in the second interlayer insulating film **14b**. The resistance element **24** is provided on the upper surface of the field insulating film **21**. One end and the other end of the resistance element **24** are electrically connected to the gate surface electrode **16a** and the gate electrode pad **19** via contact holes **23c** and **23d** penetrating the first and second interlayer insulating films **14a** and **14b**, respectively. The anode region **22a** and the cathode region **22b** of the diode **22** are formed by using low impurity concentration doped polysilicon layers that are ion-implanted with p-type impurities and n-type impurities, respectively. A doped polysilicon layer is used for the resistance element **24**, and the resistance value is controlled by the width of the doped polysilicon layer. Although the resistance element **24** is used to control the turn-on time, a wiring layer made of a doped polysilicon layer having a high impurity concentration may be arranged instead of the resistance element **24**.

(55) FIG. **21** schematically shows the connection among the gate electrode **11c** of the trench (**7a**, **7b**), the bottom dummy conductive member **9b** to the diode **22**, and the resistance element **24**. As shown in FIG. **21**, the gate electrode **11c** of the gate trench **7a** is electrically connected to the gate potential applied to the gate electrode pad **19** via the resistance element **24**. Further, the bottom dummy conductive member **9b** of the dummy trench **7b** is electrically connected to the gate potential via the resistance element **24** and the diode **22**.

(56) FIGS. **22** and **23** show the simulated turn-on waveforms and the simulated turn-off waveforms of the IGBT of the third embodiment, assuming that the collector current $I_{sub.C}$ is about 100 A. Here, in the upper graphs of these figures, for the third embodiment device, voltage and current at the gate electrode **11c** of the gate trench **7a** are depicted by the solid lines, and voltage and current at the bottom dummy conductive member **9b** of the dummy trench **7b** are depicted by the alternate long and short dash lines. Further, as a comparative example, an IGBT having a unitary dummy electrode **11d** in the dummy trench **7b** shown in FIG. **7** is used. As shown in FIG. **22**, in the third embodiment, the time change rate dv/dt of the collector voltage $V_{sub.CE}$ drops almost constantly, but in the comparative example, it is not constant and a voltage tail T is generated. In the waveforms of the gate voltage $V_{sub.GE}$ of the gate electrode **11c** in the gate trench **7a** and the gate

voltage in the comparative example, the gate voltage $V_{\text{sub.GE}}$ starts the turn-on operation from -15V . On the other hand, in the waveform of the gate voltage $V_{\text{sub.GE}}$ at the bottom dummy conductive member **9b** in the dummy trench **7b**, since the bottom dummy conductive member **9b** is connected to the diode **22**, the turn-on operation is started from 0V . As shown in FIG. **23**, as the collector voltage $V_{\text{sub.CE}}$ increases, the collector current $I_{\text{sub.C}}$ decreases. The gate voltages $V_{\text{sub.GE}}$ at the gate electrode **11c** in the gate trench **7a** of the embodiment and the gate voltage in the comparative example are both rapidly reduced after the gate voltage is cut off. On the other hand, the gate voltage $V_{\text{sub.GE}}$ at the bottom dummy conductive member **9b** in the dummy trench **7b** rises sharply in the time zone D of $0.03\ \mu\text{sec}$ to $0.07\ \mu\text{sec}$ from the time of the turn off, and then gradually decreases. Even if the gate voltage $V_{\text{sub.GE}}$ of the bottom dummy conductive member **9b** decreases, the diode **22** is in a reverse bias state and no current flows, and the potential around the dummy trench **7b** does not decrease. Since the potential around the dummy trench **7b** is maintained even when the IGBT is off, the gate current $I_{\text{sub.G}}$ hardly flows during the subsequent turn-on, and the total gate charge Q_g can be suppressed. The sharp rise in the gate voltage $V_{\text{sub.GE}}$ shown in FIG. **23** is due to the displacement current due to the sharp change in the potential around the dummy trench **7b**. It is necessary to control the impurity concentration in the anode region **22a** and the cathode region **22b** to determine the withstand voltage of the diode **22** so that the gate voltage $V_{\text{sub.GE}}$ does not become too high in the time zone D. For example, in order to suppress the gate voltage $V_{\text{sub.GE}}$ so as not to exceed 20V , the withstand voltage of the diode **22** may be set to about 35V . The other effects of the IGBT of the third embodiment are the same as those of the IGBT according to the second modification of the first embodiment.

Other Embodiments

(57) Although the present invention has been described by the embodiments described above, the statements and drawings that form part of this disclosure should not be understood as limiting the invention. It should be considered from this disclosure to those skilled in the art that various alternative embodiments, examples and operational techniques will be revealed.

(58) In the first and second embodiments, silicon (Si) has been used as the material of the semiconductor substrate, but the semiconductor material is not limited, and wide bandgap semiconductor, such as silicon carbide (SiC) or gallium nitride (GaN), may be used.

(59) As described above, the present invention includes various forms of implementations not explicitly described above, and the technical scope of the present invention is defined only by the matters specifying the invention relating to the reasonable claims from the above description. It will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the spirit or scope of the invention. Thus, it is intended that the present invention cover modifications and variations that come within the scope of the appended claims and their equivalents. In particular, it is explicitly contemplated that any part or whole of any two or more of the embodiments and their modifications described above can be combined and regarded within the scope of the present invention.

Claims

1. An insulated gate bipolar transistor, comprising: a drift layer made of semiconductor of a first conductivity type; a base region made of semiconductor of a second conductivity type on the drift layer; an emitter region made of semiconductor of the first conductivity type on the base region, an impurity concentration of the emitter region being higher than that in the drift layer; a gate electrode embedded in a gate trench through a gate insulating film, the gate trench penetrating the emitter region and the base region, the gate trench extending in a first direction in a plan view; and a dummy electrode embedded in a dummy trench through a dummy insulating film, the dummy trench penetrating the emitter region and the base region and extending in parallel with the gate trench in the first direction in a plan view, the dummy trench being disposed on each side of the

- gate trench and laterally spaced from each side of the gate trench so as to laterally face the gate trench through the base region, wherein the gate electrode is configured to be electrically connected to a gate potential, and wherein the dummy electrode includes a bottom dummy conductive member disposed at a bottom of the dummy trench such that an upper surface of the bottom dummy conductive member is located lower than a lower surface of the base region, the bottom dummy conductive member being configured to be electrically connected to the gate potential.
2. The insulated gate bipolar transistor according to claim 1, wherein the dummy electrode further includes an upper dummy conductive member that is disposed over the bottom dummy conductive member through a split insulating film, the upper dummy conductive member being configured to be electrically connected to an emitter potential.
 3. The insulated gate bipolar transistor according to claim 1, wherein the dummy trench further includes an upper embedded insulating film fillets on the bottom dummy conductive member.
 4. The insulated gate bipolar transistor according to claim 1, wherein the gate electrode in the gate trench comprises a bottom gate conductive member at a bottom of the gate trench and an upper gate conductive member that is provided over the bottom gate conductive through an insulating film, an upper surface of the bottom gate conductive member being located lower than the lower surface of the base region.
 5. The insulated gate bipolar transistor according to claim 1, further comprising a trench bottom floating layer below the base region, the trench bottom floating layer covering the bottom of the dummy trench and being made of semiconductor of the second conductivity type second conductive type, an upper surface of the trench bottom floating layer being located higher than an upper surface of the bottom dummy conductive member in the dummy trench.
 6. The insulated gate bipolar transistor according to claim 1, further comprising an accumulation layer made of semiconductor of the first conductivity type below the base region.
 7. The insulated gate bipolar transistor according to claim 1, further comprising a diode having an anode region and a cathode region, the anode region being configured to be electrically connected to the gate potential, and the cathode region being electrically connected to the bottom dummy conductive member.
 8. The insulated gate bipolar transistor according to claim 7, further comprising a resistance element, one end of the resistance element being configured to be electrically connected to the gate potential, another end of the resistance element being electrically connected to the anode region of the diode and to the gate electrode.
 9. The insulated gate bipolar transistor according to claim 1, wherein the gate trench and the dummy trench are both provided in a plurality, and the plurality of the gate trenches and the plurality of the dummy trenches are arranged in a striped shape so as to be parallel to each other in the plan view.
 10. The insulated gate bipolar transistor according to claim 1, wherein the dummy electrode further includes an upper dummy conductive member that is disposed over the bottom dummy conductive member through a split insulating film, the upper dummy conductive member being configured to be electrically connected to an emitter potential, and wherein the gate electrode in the gate trench comprises a bottom gate conductive member at a bottom of the gate trench and an upper gate conductive member that is provided over the bottom gate conductive through a split insulating film, an upper surface of the bottom gate conductive member being located lower than the lower surface of the base region, the bottom gate conductive member and the upper gate conductive member being both configured to be electrically connected to the gate potential.
-